

Use-Case Overview: UC-004 Monitor Service Health

Use-Case ID & Name	UC-004: Monitor Microservice Health & Availability
Actors	Primary: DevOps Engineer Supporting: Notification Service (External System)
Related Requirements	FR-001 (Health Pinger & Availability), FR-004 (Downtime Alerts), NFR-002 (TLS 1.2+ Security & SSO)
Description	The system periodically pings microservice health endpoints at 30 s intervals, calculates rolling 24-hr availability percentages, and dispatches multi-channel alerts upon detecting outages.

Preconditions

- DevOps Engineer is authenticated via Enterprise SSO (OAuth 2.0 / SAML).
- At least one microservice is registered in catalog with HTTPS health endpoint.
- Alert notification channels (Email, Slack, Webhook) and on-call contacts are configured.

Postconditions

- Live health status and timestamps are updated on the dashboard in real-time.
- Rolling 24-hour availability percentages are recalculated and displayed.
- Outage incidents are logged and notifications dispatched to on-call contacts.

Main Success Scenario

Step	Actor / System	Action / Event Description
1	DevOps Engineer	Navigates to the Health Monitoring dashboard and selects a microservice (or group) to monitor.
2	System	Validates the engineer's SSO session. «include» Authenticate via Enterprise SSO
3	System	Retrieves the health-check endpoint URL and configured ping interval (default 30 s) from the catalog.
4	System	Dispatches an HTTPS health probe (GET request over TLS 1.2+) to the microservice endpoint.
5	System	Receives a 200 OK response within the timeout window (5 s) and records probe as Healthy .
6	System	Updates the service status on the portal to Healthy with last-checked timestamp.
7	System	Recalculates and displays the rolling 24-hour service availability percentage.
8	System	Waits for the configured interval (30 s) and loops back to Step 4 for the next probe cycle.
9	DevOps Engineer	Views real-time health metrics, uptime availability trends, and historical logs on the dashboard.

Alternate Flow: Outage Detected (3 Consecutive Failed Probes)

Step	Actor / System	Action / Event Description
5a	System	Probe fails (non-200 status, timeout, or connection reset). System records failure and increments counter.
5b	System	Repeats probe at next interval. If successful, resets failure counter and resumes Step 6 of Main Scenario.
5c	System	Upon 3 consecutive failures, marks service as Down and logs outage incident with initial failure timestamp.
5d	System	«extend» Dispatch Downtime Alert Dispatches alert payload to configured channels (Email, Slack, Webhook).
5e	Notification Svc	Delivers real-time alert notifications to designated on-call contacts.
5f	DevOps Engineer	Receives alert, accesses dashboard, and analyzes service dependencies, error logs, and blast radius.
5g	System	Continues probing; upon receiving 200 OK, marks service Healthy , calculates downtime, and sends Recovery alert.